**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# LESSON 113 · Lesson 113 · Open Source and Closed Source Decisions

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

## Do Now · Recall previous learning

Q1. Name a utility for recovering files after hardware loss.

Q2. What is the effect of enqueue(X), enqueue(Y), dequeue() on an initially empty FIFO queue?

Q3. What is the time complexity of sequential search in the worst case?

Q4. After pushing A then B onto a stack, which is popped first?

**Do Now** Recall previous learning | **I Do** Read new knowledge | **We Do** Practise with guidance | **You Do** Work independently | **Review & Improve** Check understanding

BLACK PEN Answer all questions first. GREEN PEN Then: tick and rewrite mistakes.

# I Do · Read new knowledge

## Key vocabulary

open source — Code distributed under a licence allowing inspection and specified reuse.

closed source — Source and modification rights normally controlled by owner.

licence — Legal conditions attached to software use.

vendor lock-in — Difficult or expensive dependence on one supplier.

## New learning

Source availability — Open-source licences allow access to source code with specified reuse rights.

Control vs support — Closed-source vendors control release; support and integration vary by product.

Make a justified choice — Evaluate total cost, auditability, maintenance, expertise and obligations.

## Worked example

A council wants to adapt an accessible web service using its development team. Explain an open-source advantage and a risk.

1. Model answer — Relevant open-source licence can permit modification and inspection, allowing accessibility changes. Risk: the council must ensure support, security patching and licence compliance; source availability alone guarantees neither quality nor safety. — Check: link each claim to the scenario and use precise OCR terminology.

# We Do · Practise with guidance

A small medical practice needs guaranteed vendor response and has no developers. Compare closed-source options.

What must we decide?

Identify the inputs, relevant facts and exact output or judgement required.

Work through the method

Use the middle stage of the visual to follow the process and state each intermediate result.

Shared worked answer

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

A proprietary supported package may offer a service contract and managed upgrades; assess data export, recurring licensing, vendor lock-in and contract quality. Source model alone does not decide security.

BLACK PEN Answer all questions first. GREEN PEN Then: tick and rewrite mistakes.

## You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

### Question 1

Define open-source and closed-source software and state a licensing implication. [6]

### Question 2

Explain why free-of-charge software is not necessarily open source. [4]

### Question 3

Evaluate the choice for a charity's public website considering skills, privacy, costs and vendor dependence. [8]

## Exit Ticket · Answer before checking

Open-source software is automatically more secure than closed-source software.





BLACK PEN Answer all questions first. GREEN PEN Then: tick and rewrite mistakes.

## Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

### Page 1 onwards · Do Now

Q1. Backup and restore utility.

Q2. X is removed; Y remains at the front.

Q3. O(n).

Q4. B, because a stack is LIFO.

### You Do pages · Independent questions

Q1. Open source provides source and rights under its licence [2]; closed source restricts source/modification [2]; licensing conditions matter [2].

Q2. Price and source-code rights differ [2]; a free proprietary program can withhold code/modification rights [2].

Q3. Modification/audit potential [2], support skills/security updates [2], licence/TCO [2], justified conclusion for circumstances [2].

### Exit Ticket · Model answer

False. Security depends on implementation, maintenance and context.

## Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

**1. Copy the whole original question:**

**2. Write a complete new answer in GREEN PEN. Show working if needed:**